

Reliable Normal Projection with Healthy-Reference Calibration for Brain MRI Anomaly Localization

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Abstract. Amyotrophic Lateral Sclerosis (ALS) exposes a central difficulty for brain MRI anomaly localization: clinically important brainstem involvement affects bulbar and respiratory function, but routine MRI provides no voxel-wise disease ground truth. We study this setting as a reliability problem for normal-projection models. *Reliable Normal Projection with Healthy-Reference Calibration* maps an unseen scan toward a learned healthy reference and converts patient-to-healthy deviations into calibrated selective evidence. Because residuals may reflect pathology, anatomical variability, artifacts, registration mismatch, or projection instability, we combine multi-view residual cues, projection-disagreement uncertainty, and false-positive-controlled calibration on held-out healthy scans before patient evaluation. The output is calibrated evidence, not a supervised segmentation mask or voxel-wise ALS pathology map. We evaluate complementary regimes spanning visible lesion localization, mask-free neurodegeneration analysis, and ALS brainstem ROI assessment. The framework provides a reliability-oriented protocol for evaluating normal-projection medical imaging models under severe annotation scarcity without threshold tuning on abnormal data.

Keywords: Medical Foundation Model Evaluation · Brain MRI · Normal Projection · Healthy-Reference Calibration · Annotation Scarcity

1 Introduction

Amyotrophic Lateral Sclerosis (ALS) is a progressive neurodegenerative disease characterized by motor-neuron degeneration, progressive weakness, disability, and life-threatening respiratory impairment [7, 11, 15]. Bulbar and respiratory dysfunction affect speech, swallowing, airway protection, ventilation, prognosis, and survival, making the brainstem a clinically meaningful target for magnetic resonance imaging (MRI)-based analysis. However, routine *in-vivo* ALS MRI does not provide visually apparent voxel-wise disease boundaries, and ALS-related brainstem involvement is subtle, low-contrast, and difficult to separate from normal anatomical variability, partial-volume effects, imaging noise, and

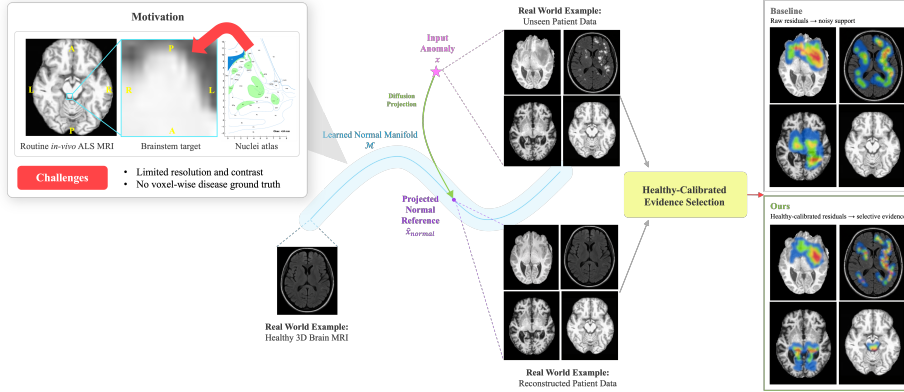


Fig. 1: Motivation and overview. Routine *in-vivo* ALS MRI lacks established voxel-wise ground truth for brainstem involvement. Our framework maps unseen MRI toward a learned healthy manifold and converts residual deviations into healthy-calibrated selective evidence for reliability-oriented evaluation.

registration uncertainty. Anatomical nuclei atlases can support region-of-interest interpretation, but they do not constitute voxel-wise disease ground truth; Figure 1 summarizes this motivation.

This setting reflects a broader challenge in annotation-scarce medical imaging. Visible lesion benchmarks allow mask-based localization assessment, whereas ALS brainstem MRI requires anatomically interpretable evidence of deviation from healthy anatomy without voxel-wise disease masks or abnormal-data threshold tuning. Normal-only anomaly localization naturally fits this setting, using reconstruction, feature-distribution, and diffusion-based normal-projection models to learn healthy anatomy and detect test-time deviations [2, 3, 9, 12, 16, 21, 28, 30, 32–36]. However, raw patient-to-healthy residuals are not reliable anomaly evidence by default: large deviations may reflect pathology, but may also arise from benign anatomical variation, artifacts, imperfect registration, scanner or protocol differences, or projection instability.

We address this limitation with *Reliable Normal Projection with Healthy-Reference Calibration*. Rather than treating residual magnitude as an anomaly map, we convert patient-to-healthy deviations into calibrated selective evidence. The framework is trained only on healthy MRI. At inference, an unseen scan is projected toward a healthy reference; complementary deviations are extracted across image-space, latent-space, and high-frequency views; and projection-disagreement uncertainty estimates whether residual evidence is stable across stochastic normal projections. Held-out healthy scans calibrate residual fusion, uncertainty filtering, and false-positive-controlled evidence selection before downstream patient evaluation. Thus, the final map represents projection-consistent evidence selected according to healthy-reference false-positive behavior, not a supervised disease segmentation mask or a voxel-wise ALS pathology map.

To evaluate this ALS-driven problem fairly, we use complementary regimes rather than relying on ALS data alone. Visible lesion benchmarks assess mask-based spatial localization; mask-free neurodegeneration settings assess anatomical evidence concentration; and ALS brainstem analysis serves as an ROI-centric reliability stress test under annotation scarcity. In ALS, we evaluate whether calibrated evidence becomes more specific to clinically motivated brainstem regions without claiming voxel-wise disease segmentation or clinical validation. We further compare the task-oriented ALS ROI against unrelated size-matched control ROIs to reduce the possibility that ROI concentration is explained solely by brainstem-focused cropping. Throughout the evaluation, patient labels, lesion masks, and disease-specific regions are used only for evaluation, while model fitting and calibration use healthy reference data.

Our contributions are:

- We formulate ALS-motivated brain MRI anomaly localization as healthy-reference calibrated evidence evaluation under severe annotation scarcity, rather than supervised disease segmentation.
- We propose a normal-projection framework that converts patient-to-healthy deviations into calibrated selective evidence using multi-view residuals, projection-disagreement uncertainty, and held-out healthy calibration.
- We evaluate the framework across visible lesion, mask-free neurodegeneration, and ALS brainstem ROI regimes, including unrelated ROI controls for the ALS setting.

2 Related Work

2.1 Normal-only anomaly localization

Normal-only anomaly localization is widely used when voxel-wise pathology labels are unavailable, learning healthy anatomical regularities and detecting deviations from a reconstructed or projected normal reference. Representative approaches include adversarial reconstruction methods such as AnoGAN and f-AnoGAN [28, 29], autoencoder- and VAE-based models [2, 3, 21, 27], context-encoding VAEs [36], and feature-distribution methods such as PaDiM [9]. These methods reduce dependence on dense annotations, but residual or feature-deviation maps may also reflect anatomical variability, scanner effects, registration mismatch, skull-stripping errors, or partial-volume artifacts. Thus, our work focuses on reliability-oriented calibration of patient-to-healthy evidence rather than direct residual thresholding or supervised segmentation.

2.2 Diffusion-based normal projection

Diffusion models provide expressive generative priors for denoising, reconstruction, inpainting, and normal projection [16, 19, 26, 30, 31]. They have been widely adopted for medical anomaly detection, including DDPM-based methods [33],

AnoDDPM [35], fast and patched diffusion models [4, 25], Masked Bernoulli Diffusion [32], MDPS [34], and ANDi [12]. However, plausible healthy projections can still produce residuals driven by sampling variability, projection ambiguity, preprocessing mismatch, age-related morphology, scanner differences, or benign anatomical variation. Our framework is complementary to these backbones: it treats the projected reference as a starting point and selects residual deviations that are stable, low-uncertainty, and compatible with healthy-reference false-positive behavior.

2.3 Reliability under annotation scarcity

Uncertainty estimation and calibration are widely used to quantify ambiguity and improve reliability, including Monte Carlo dropout, predictive uncertainty modeling, and deep ensembles [13, 20, 22], with calibration especially important under domain mismatch and acquisition variability [14, 23]. Here, the output is neither a class probability nor a supervised segmentation mask; uncertainty instead evaluates whether normal-projection residual evidence remains stable across stochastic projections.

This reliability-oriented formulation is motivated by annotation-scarce evaluation. Visible lesion datasets allow overlap-based assessment, whereas neurodegenerative diseases often require anatomical rather than voxel-wise interpretation. In ALS, clinically relevant brainstem nuclei are involved in bulbar and respiratory function, yet routine MRI provides no established voxel-wise disease ground truth. ROI-centric evidence concentration can assess anatomical plausibility, but not voxel-wise disease correctness or clinical utility. Accordingly, we treat ALS brainstem analysis as an ROI-centric reliability stress test and position healthy-reference calibrated normal projection as an evaluation framework for annotation-scarce brain MRI, rather than as a general-purpose medical foundation model or clinically validated ALS biomarker [6, 24].

3 Problem Setting and Benchmark Regimes

3.1 ALS-driven annotation-scarce localization

Our target application is brainstem analysis in ALS, where bulbar and respiratory dysfunction affect speech, swallowing, airway protection, ventilation, prognosis, and survival, and where the brainstem contains functionally relevant nuclei and pathways [7, 11, 15]. However, ALS-related brainstem involvement has no established voxel-wise disease ground truth in routine *in-vivo* MRI; abnormalities are subtle, low-contrast, and difficult to separate from normal anatomical variability, imaging noise, partial-volume effects, and registration uncertainty. Anatomical atlases can support region-of-interest (ROI) interpretation, but they should not be treated as voxel-wise disease labels [5, 17].

We therefore formulate the task as *annotation-scarce anomaly evidence localization* rather than supervised disease segmentation. Given an unseen patient

scan x , the goal is to produce a continuous evidence map $S(x)$ indicating calibrated deviation from healthy reference anatomy, interpreted through anatomical ROIs and control regions when voxel-wise disease masks are unavailable. Thus, $S(x)$ is calibrated evidence, not a voxel-wise ALS pathology map or clinically validated biomarker.

3.2 Healthy-only learning and calibration protocol

Let $\mathcal{X}_H$ denote healthy reference MRI scans and $\mathcal{X}_P$ downstream patient cohorts. The model is trained only on $\mathcal{X}_H$; all model parameters, fusion weights, uncertainty thresholds, and score thresholds are fixed before patient evaluation. No downstream lesion mask, diagnostic label, or disease-specific ROI is used for fitting, hyperparameter selection, residual fusion, uncertainty filtering, or calibration; downstream references are used only for evaluation.

OpenBHB [10] provides the healthy reference data and is split into non-overlapping training, validation, calibration, and healthy-test subsets using a 70/10/10/10 ratio. Training learns the normal-projection backbone; validation selects fusion and uncertainty-penalty parameters; calibration fixes the uncertainty cutoff and FPR-controlled score threshold; and healthy-test scans are reserved for final false-positive analysis. This separation is necessary because patient-to-healthy residuals may reflect pathology, but also anatomical variability, artifacts, registration mismatch, scanner or protocol differences, partial-volume effects, or projection instability. Healthy-reference calibration therefore selects evidence according to held-out healthy false-positive behavior before downstream patient evaluation.

3.3 Benchmark regimes and evaluation references

Because ALS brainstem data lack voxel-wise disease masks, we use complementary regimes: BraTS [1] and MSLesSeg [8] assess visible lesion localization and small-lesion sensitivity with voxel-wise masks; ADNI [18] supports mask-free neurodegeneration analysis through ROI-centric anatomical evidence concentration; and PULSE supports brainstem-focused assessment of evidence specificity and off-target reduction.

These regimes are not identical tasks: visible lesion datasets test focal abnormality localization, whereas ADNI and PULSE test whether the same healthy-calibrated framework produces anatomically meaningful, low-uncertainty evidence when disease effects are diffuse or voxel-wise boundaries are unavailable.

Accordingly, ALS brainstem analysis is treated as an ROI-centric reliability stress test under annotation scarcity, not as clinical validation of ALS-specific pathology localization. For PULSE, the atlas-derived brainstem mask defines the focused field of view, whereas ROI-centric metrics use task-oriented nuclei or pathway regions within that mask. To test whether ROI concentration is driven only by brainstem cropping, we compare the task-oriented ALS ROI with $J = 100$ size-matched unrelated control ROIs sampled inside the brainstem mask while excluding the target ROI neighborhood.

3.4 Imaging inputs and evaluation principle

All datasets are brought into a common spatial and intensity framework before training or evaluation. T₁-weighted MRI (T1w) is the primary structural modality because it is consistently available across cohorts, although it may be insufficient for subtle microstructural ALS-related changes. All T1w volumes are registered to a common stereotaxic space, resampled, and intensity-normalized; voxel-wise masks are transformed consistently and used only for evaluation.

For PULSE, we evaluate whole-brain T1w, brainstem-cropped T1w, and exploratory brainstem-cropped T1w+fractional anisotropy (FA) inference. Whole-brain inference tests broad-field specificity, whereas brainstem cropping restricts inference to the clinically motivated region and reduces off-target residuals. For T1w+FA, same-subject FA maps are registered into T1w space, concatenated as an additional input channel, and evaluated using the same healthy-reference calibration protocol as an exploratory microstructural complement rather than independent clinical validation.

Overall, the resulting maps are evaluated as calibrated selective evidence under annotation scarcity, combining mask-based localization, ROI-centric anatomical plausibility, unrelated ROI controls, and held-out healthy false-positive control.

4 Reliable Normal Projection with Healthy-Reference Calibration

Figure 2 summarizes the proposed healthy-only normal-projection framework. Unlike methods that directly threshold patient-to-healthy residuals, we treat residuals as candidate evidence whose reliability must be estimated before interpretation. Our contribution is a reliability-oriented calibration procedure that converts multi-view residuals into projection-consistent, false-positive-controlled selective evidence using only healthy reference scans, targeting annotation-scarce clinical settings where patient labels, lesion masks, or disease-specific ROIs cannot tune anomaly thresholds.

4.1 Healthy-only normal projection backbone

Let $\mathcal{X}_H$ denote healthy reference volumes $x_h \in \mathbb{R}^{H \times W \times D}$, with no pathology label or lesion mask. We learn a compact normal-manifold representation using an encoder-decoder pair (E_ϕ, D_ψ) :

$$z_h = E_\phi(x_h), \quad \tilde{x}_h = D_\psi(z_h), \quad (1)$$

where z_h is the latent representation and $\tilde{x}_h$ the reconstructed healthy image. A latent diffusion teacher T_θ is trained on healthy latent codes [16, 19, 26, 31], and its normal projection is distilled into a fast student S_ω :

$$\hat{z}_{\text{normal}} = S_\omega(z). \quad (2)$$

The full backbone is trained only on healthy MRI, without patient labels, lesion masks, diagnostic labels, or disease-specific ROIs.

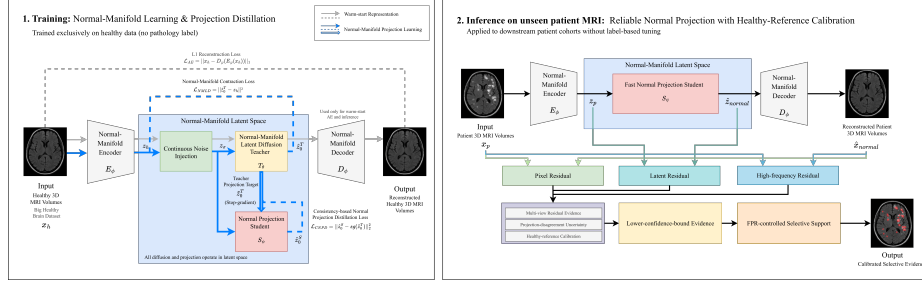


Fig. 2: Reliable normal projection with healthy-reference calibration. Healthy-only training builds the latent normal-projection backbone; inference maps an unseen scan to a healthy reference and converts multi-view deviations into uncertainty-aware, FPR-controlled selective evidence. The output is calibrated evidence, not a supervised segmentation mask or voxel-wise ALS pathology map.

4.2 Patient-to-healthy residual evidence

At inference, an unseen scan x_p is encoded and projected toward the learned healthy manifold:

$$z_p = E_\phi(x_p), \quad \hat{z}_{normal} = S_\omega(z_p), \quad \hat{x}_{normal} = D_\psi(\hat{z}_{normal}). \quad (3)$$

A naive detector would threshold $|x_p - \hat{x}_{normal}|$, but this residual may also arise from anatomical variability, artifacts, registration mismatch, scanner or protocol differences, partial-volume effects, or projection instability. We therefore extract complementary image-space, latent-space, and high-frequency residual views:

$$\begin{aligned} R_{px} &= |x_p - \hat{x}_{normal}|, \\ R_{lat} &= \uparrow(\text{Red}_c(|z_p - \hat{z}_{normal}|)), \\ R_{hf} &= |\mathcal{H}(x_p) - \mathcal{H}(\hat{x}_{normal})|, \end{aligned} \quad (4)$$

where $\text{Red}_c(\cdot)$ is the channel-wise mean of absolute latent differences, $\uparrow(\cdot)$ is trilinear interpolation to the image grid, and $\mathcal{H}(\cdot)$ is a fixed non-learned 3D Laplacian high-pass operator.

Let $\mathcal{M} = \{\text{px}, \text{lat}, \text{hf}\}$ denote residual views and $\mathcal{S} = \{1, 1/2, 1/4\}$ spatial scales. For $m \in \mathcal{M}$ and $s \in \mathcal{S}$, residual maps are fused as

$$R_{\text{fuse}} = \sum_{m \in \mathcal{M}} \sum_{s \in \mathcal{S}} w_{m,s} \uparrow(\mathcal{N}(R_m^{(s)})), \quad (5)$$

where $\mathcal{N}(\cdot)$ denotes normalization and $w_{m,s}$ are selected using only the held-out healthy validation split and fixed before downstream patient evaluation.

4.3 Projection-disagreement uncertainty

Residual magnitude alone does not indicate reliability, so we estimate projection-disagreement uncertainty from K stochastic normal projections while keeping

the trained model and preprocessing fixed. Each projection produces $R_{\text{fuse}}^{(k)}$, and for each voxel $v \in \Omega$,

$$\bar{R}(v) = \frac{1}{K} \sum_{k=1}^K R_{\text{fuse}}^{(k)}(v), \quad U(v) = \text{Std}_k \left(R_{\text{fuse}}^{(k)}(v) \right). \quad (6)$$

Here, $\bar{R}(v)$ is mean residual evidence and $U(v)$ measures projection disagreement. Stable deviations across plausible normal projections are treated as more reliable than residuals depending on a single stochastic sample; we use $K = 4$ as the default trade-off between uncertainty stability and computational cost.

4.4 Healthy-reference calibrated selective evidence

Rather than thresholding $\bar{R}(v)$, we construct a lower-confidence-bound evidence score:

$$S_{\text{LCB}}(v) = [\bar{R}(v) - \lambda_U U(v)]_+, \quad (7)$$

where $[\cdot]_+$ denotes the positive part and λ_U is selected using only the healthy validation split. The final operating point is calibrated on a separate healthy calibration split: given target FPR α , $q_{1-\alpha}$ is computed from the empirical $(1-\alpha)$ -quantile of scan-level maximum LCB scores, and τ_U from the healthy calibration uncertainty distribution.

The FPR-controlled selective support and final evidence map are

$$\begin{aligned} \mathcal{A}_\alpha &= \{v \in \Omega : S_{\text{LCB}}(v) > q_{1-\alpha}, U(v) \leq \tau_U\}, \\ S_{\text{final}}(v) &= S_{\text{LCB}}(v) \cdot \mathbf{1}[v \in \mathcal{A}_\alpha]. \end{aligned} \quad (8)$$

Fusion weights, uncertainty penalties, uncertainty filtering, and score thresholds are all fixed before downstream patient evaluation.

The output S_{final} is not a supervised segmentation mask. In visible lesion datasets, it is compared with voxel-wise pathology masks for localization assessment; in mask-free neurodegeneration and ALS brainstem analysis, it is interpreted as anatomically grounded deviation evidence, not as voxel-wise ALS pathology or clinical disease validation.

5 Experimental Protocol

5.1 Training protocol and fair comparison

We evaluate healthy-reference calibrated normal projection under a strict healthy-only paradigm: OpenBHB [10] is used for reference learning, validation, calibration, and false-positive analysis, whereas all downstream cohorts and annotations are reserved for evaluation. OpenBHB contains 5,330 healthy samples with mean age 25.2 years and range 5–86 years, including approximately 1,000 samples aged 60–80; because it remains younger-skewed and lacks subjects beyond the eighties, we use class-balanced loss weighting with batch-wise normalization.

Healthy scans are split into training, validation, calibration, and healthy-test subsets using a 70/10/10/10 ratio: training fits the model, validation selects checkpoints and fusion/uncertainty parameters, calibration fixes uncertainty and FPR thresholds, and healthy-test scans are reserved for final false-positive analysis. No abnormal scans from BraTS [1], MSLesSeg [8], ADNI [18], or PULSE are used during training, validation, or calibration.

All methods use identical preprocessing, including spatial registration, re-sampling, masking, and intensity normalization; registered outputs are visually checked, with no scan excluded after preprocessing. Baselines are trained or fitted on the same healthy split whenever applicable, use original hyperparameters without downstream tuning, and share the same healthy calibration protocol for thresholded analyses. Our framework follows three healthy-only stages: normal-manifold learning, latent diffusion teacher training, and fast projection student distillation; all experiments use $n = 5$ fixed-split, fixed-seed runs in `Python 3.10` and `PyTorch 2.5` on an NVIDIA A100 GPU.

5.2 Evaluation cohorts and baselines

We evaluate four regimes: BraTS [1] and MSLesSeg [8] for visible lesion localization, ADNI [18] for mask-free neurodegeneration analysis, and PULSE for ALS brainstem-focused ROI assessment. PULSE has mean age 69.26 years, male-to-female ratio 0.635:0.365, and mean disease duration 21.42 months with values up to 134 months after clinical metadata quality control; ALSFRS-R, SNIP, and NIV status are available, but clinical-association analysis is left for future longitudinal validation.

For PULSE, we evaluate whole-brain T1w, brainstem-cropped T1w, and exploratory brainstem-cropped T1w+FA inference; the atlas-derived brainstem mask defines the focused field of view, while the task-oriented ALS ROI is only an anatomical evaluation reference. For T1w+FA, same-subject FA maps are registered into T1w space, concatenated as an additional channel, and evaluated with the same calibration protocol as an exploratory microstructural complement, not independent clinical validation.

We compare against healthy-only baselines: AE-UNet [27], VAE-UNet [27], ceVAE [36], f-AnoGAN [28], PaDiM [9], DDPM [16], AnoDDPM [35], DDIM [30], Masked Bernoulli Diffusion [32], MDPS [34], and ANDi [12]; internal variants isolate fusion, uncertainty, LCB, and calibration components.

5.3 Metrics and reporting protocol

For voxel-wise lesion datasets, we report Pixel-AUPRC, Max-IoU, Recall@10%, small-lesion Recall@5%, and Top-1% Dice. Pixel-AUPRC measures voxel-level ranking, Recall@ p measures lesion retrieval among the top $p\%$ scores, and Max-IoU is only a threshold-swept reference overlap metric, not used for calibration or threshold selection. For FPR-controlled LCB analysis, we vary target healthy FPR α and report lesion recall, small-lesion recall, and held-out OpenBHB healthy-test FPR.

Table 1: Main visible lesion localization on BraTS and MSLesSeg. Comparison with representative normal-only baselines.

Method	BraTS [1]			MSLesSeg [8]		
	Max-IoU $\uparrow$	Pixel-AUPRC $\uparrow$	Recall@10% $\uparrow$	Max-IoU $\uparrow$	Pixel-AUPRC $\uparrow$	Recall@10% $\uparrow$
AE-UNet [27]	0.27 $\pm$ 0.01	0.40 $\pm$ 0.01	0.44 $\pm$ 0.02	0.15 $\pm$ 0.02	0.23 $\pm$ 0.01	0.40 $\pm$ 0.02
PaDiM [9]	0.29 $\pm$ 0.02	0.46 $\pm$ 0.01	0.48 $\pm$ 0.02	0.20 $\pm$ 0.01	0.29 $\pm$ 0.02	0.48 $\pm$ 0.01
DDPM [16]	0.28 $\pm$ 0.02	0.45 $\pm$ 0.02	0.49 $\pm$ 0.02	0.19 $\pm$ 0.02	0.30 $\pm$ 0.02	0.47 $\pm$ 0.02
MDPS [34]	0.37 $\pm$ 0.01	0.50 $\pm$ 0.02	0.56 $\pm$ 0.02	0.26 $\pm$ 0.02	0.37 $\pm$ 0.01	0.52 $\pm$ 0.01
ANDi [12]	0.39 $\pm$ 0.01	0.52 $\pm$ 0.02	0.58 $\pm$ 0.01	0.28 $\pm$ 0.02	0.39 $\pm$ 0.03	0.54 $\pm$ 0.02
Ours – Multiscale Only	0.41 $\pm$ 0.02	0.55 $\pm$ 0.01	0.62 $\pm$ 0.01	0.30 $\pm$ 0.01	0.42 $\pm$ 0.01	0.57 $\pm$ 0.02
Ours – Full Model	0.46 $\pm$ 0.01	0.61 $\pm$ 0.01	0.69 $\pm$ 0.02	0.35 $\pm$ 0.01	0.48 $\pm$ 0.02	0.65 $\pm$ 0.02

For ADNI and PULSE, which lack voxel-wise disease masks, we report ROI concentration, reliable ROI concentration within $\mathcal{A}_\alpha$, FP@Non-ROI, and Score IQR; for ALS, we add an unrelated ROI control with $J = 100$ size-matched random ROIs sampled inside the brainstem mask, excluding the target ROI neighborhood. Robustness is tested under noise, intensity perturbation, mild spatial mismatch, and smoothing at the same calibrated operating point; results are mean values with 95% confidence intervals over $n = 5$ runs, with calibration choices fixed before downstream evaluation.

6 Results

Under a strict healthy-only protocol, we evaluate calibrated selective evidence across visible lesion localization, FPR-controlled small-lesion sensitivity, mask-free ROI concentration, ALS ROI controls in PULSE, and reliability ablations. Unless otherwise stated, values are mean $\pm$ 95% confidence interval over $n = 5$ runs, with all parameters and thresholds fixed before downstream evaluation.

6.1 Visible lesion localization

We evaluate BraTS [1] and MSLesSeg [8], where voxel-wise masks support localization assessment; Table 1 shows that the full model performs best in thresholded overlap, voxel-level ranking, and high-confidence lesion retrieval. Max-IoU is a threshold-swept reference overlap metric, not used for calibration, operating-point selection, or mask-free evaluation.

On BraTS, ours improves over ANDi [12] from 0.39 to 0.46 in Max-IoU, 0.52 to 0.61 in Pixel-AUPRC, and 0.58 to 0.69 in Recall@10%; on MSLesSeg, gains are 0.28 to 0.35, 0.39 to 0.48, and 0.54 to 0.65. This shows that healthy-calibrated selective evidence remains effective in visible lesion regimes, while not validating subtle neurodegenerative or ALS-specific abnormalities by itself.

The multiscale-only variant improves over external baselines, but the full model remains strongest, indicating that multiscale residual evidence alone does not account for the observed gains. Section 6.4 further isolates the effects of uncertainty-aware scoring and healthy-reference calibration.

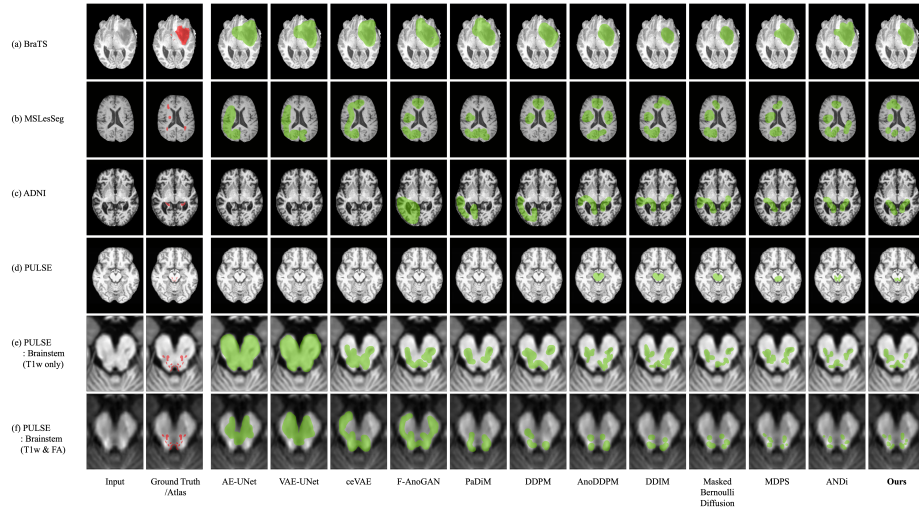


Fig. 3: Qualitative results across benchmark regimes. Red denotes lesion masks or ROI references; green denotes calibrated selected evidence. Examples illustrate selective evidence localization across visible lesion and mask-free ROI-centric settings.

6.2 Qualitative behavior and small-lesion sensitivity

Figure 3 shows examples across visible lesion and ROI-centric regimes. BraTS [1] and MSLesSeg [8] overlays are compared with voxel-wise masks, whereas ADNI [18] and PULSE are interpreted through anatomical ROI references. Calibrated evidence is more compact than residual-based or diffusion-based supports while remaining concentrated around lesions or relevant ROIs, suggesting off-target suppression without empty-map collapse.

Because FPR-controlled calibration could suppress weak abnormalities, we evaluate small-lesion subsets: BraTS lesions below 5,000 voxels and MSLesSeg lesions below 300 voxels. Table 2 shows that the full model improves over ANDi [12] across all small-lesion metrics. On BraTS, Recall@5%, Top-1% Dice, and calibrated recall improve from 0.49, 0.19, and 0.45 to 0.62, 0.31, and 0.60; on MSLesSeg, they improve from 0.41, 0.15, and 0.36 to 0.56, 0.26, and 0.53. Thus, calibrated selective support preserves compact lesion evidence while filtering unstable residual responses.

Table 3 analyzes the operating-point trade-off. Uncalibrated LCB support gives the highest lesion recall but high healthy-test FPR 0.142. At $\alpha = 0.05$, healthy-test FPR decreases to 0.047 while retaining most sensitivity: BraTS recall 0.67, BraTS small-lesion recall 0.60, MSLesSeg recall 0.63, and MS small-lesion recall 0.53. This FPR reflects OpenBHB healthy-reference false-positive control, not a guaranteed downstream FPR under every age, scanner, or site distribution.

Table 2: Small-lesion localization on BraTS and MSLesSeg. Small lesions are defined as BraTS < 5,000 voxels and MSLesSeg < 300 voxels; values are mean $\pm$ 95% confidence interval over $n = 5$ runs.

Method	BraTS small lesions [1]				MSLesSeg small lesions [8]			
	Recall@5% $\uparrow$	Top-1% Dice $\uparrow$	Calibrated recall $\uparrow$		Recall@5% $\uparrow$	Top-1% Dice $\uparrow$	Calibrated recall $\uparrow$	
AE-UNet [27]	0.38 $\pm$ 0.02	0.12 $\pm$ 0.01	0.34 $\pm$ 0.02		0.29 $\pm$ 0.02	0.08 $\pm$ 0.01	0.25 $\pm$ 0.02	
DDPM [16]	0.44 $\pm$ 0.01	0.17 $\pm$ 0.01	0.39 $\pm$ 0.02		0.38 $\pm$ 0.02	0.12 $\pm$ 0.01	0.32 $\pm$ 0.02	
MDPS [34]	0.48 $\pm$ 0.01	0.19 $\pm$ 0.02	0.43 $\pm$ 0.02		0.40 $\pm$ 0.02	0.14 $\pm$ 0.01	0.35 $\pm$ 0.02	
ANDi [12]	0.49 $\pm$ 0.01	0.19 $\pm$ 0.03	0.45 $\pm$ 0.02		0.41 $\pm$ 0.01	0.15 $\pm$ 0.01	0.36 $\pm$ 0.02	
Ours – Full Model	0.62 $\pm$ 0.01	0.31 $\pm$ 0.02	0.60 $\pm$ 0.02		0.56 $\pm$ 0.01	0.26 $\pm$ 0.01	0.53 $\pm$ 0.02	

6.3 ROI-centric neurodegeneration and ALS brainstem assessment

ADNI [18] and PULSE are mask-free regimes; evidence is evaluated using ROI concentration and off-target suppression rather than voxel-wise disease segmentation. Table 4 shows that the method improves ROI metrics and reduces off-target support. On ADNI, it improves over ANDi [12] from 0.63 to 0.72 in ROI concentration and 0.57 to 0.66 in reliable ROI concentration, while reducing FP@Non-ROI from 0.14 to 0.09, suggesting plausible anatomical concentration rather than global suppression or voxel-wise neurodegeneration segmentation.

For PULSE, whole-brain T1w improves ROI concentration over ANDi from 0.42 to 0.53, but permits off-target support; brainstem cropping increases ROI concentration to 0.62, reliable ROI concentration to 0.57, and reduces FP@Non-ROI to 0.18. Exploratory T1w+FA improves these values to 0.65, 0.60, and 0.16, with Score IQR decreasing from 0.34 to 0.30 and 0.29; as FA is diffusion-derived and the analysis remains ROI-centric, this is interpreted as an exploratory microstructural complement rather than independent clinical validation.

To test whether the ALS ROI concentration observed in PULSE is driven solely by brainstem-focused cropping, we use $J = 100$ size-matched unrelated ROIs sampled inside the brainstem mask while excluding the target ROI neighborhood. In the brainstem T1w setting, the task-oriented ALS ROI reaches 0.62 versus 0.20 ± 0.06 for controls, giving $3.10\times$ enrichment with permutation $p < 0.05$. This suggests that selected evidence is not uniformly distributed across arbitrary brainstem regions, but remains anatomical plausibility evidence; overall, the PULSE analysis is treated as an ROI-centric reliability stress test producing calibrated, hypothesis-generating evidence rather than clinical validation of ALS-specific localization.

Table 3: Target FPR versus lesion sensitivity under FPR-controlled LCB localization. Rows vary target FPR α and report retained sensitivity. Thresholds are healthy-calibrated; healthy-test FPR is measured on held-out OpenBHB scans.

Operating point	Target FPR α	BraTS recall $\uparrow$	BraTS small recall $\uparrow$	MSLesSeg recall $\uparrow$	MS small recall $\uparrow$	Healthy-test FPR $\downarrow$
Uncalibrated LCB support	–	0.73 $\pm$ 0.02	0.65 $\pm$ 0.02	0.68 $\pm$ 0.02	0.59 $\pm$ 0.02	0.142 $\pm$ 0.021
FPR-controlled LCB	0.100	0.70 $\pm$ 0.02	0.63 $\pm$ 0.02	0.66 $\pm$ 0.02	0.57 $\pm$ 0.02	0.092 $\pm$ 0.018
FPR-controlled LCB	0.050	0.67 $\pm$ 0.02	0.60 $\pm$ 0.02	0.63 $\pm$ 0.02	0.53 $\pm$ 0.02	0.047 $\pm$ 0.012
FPR-controlled LCB	0.010	0.61 $\pm$ 0.02	0.54 $\pm$ 0.02	0.57 $\pm$ 0.02	0.47 $\pm$ 0.02	0.012 $\pm$ 0.006
FPR-controlled LCB	0.005	0.56 $\pm$ 0.03	0.48 $\pm$ 0.02	0.52 $\pm$ 0.02	0.41 $\pm$ 0.02	0.006 $\pm$ 0.004

Table 4: ROI-centric selective evidence on ADNI and PULSE. ADNI uses hippocampal ROIs; PULSE includes whole-brain T1w, brainstem T1w, and brainstem T1w+FA regimes. Values are mean $\pm$ 95% confidence interval over $n = 5$ runs.

Dataset / regime	Method	ROI conc. $\uparrow$	Reliable ROI conc. $\uparrow$	FP@Non-ROI $\downarrow$	Score IQR $\downarrow$
ADNI [18]	AE-UNet [27]	0.51 ± 0.01	0.45 ± 0.02	0.25 ± 0.01	0.28 ± 0.01
ADNI [18]	DDPM [16]	0.56 ± 0.01	0.50 ± 0.02	0.18 ± 0.01	0.27 ± 0.01
ADNI [18]	ANDi [12]	0.63 ± 0.01	0.57 ± 0.02	0.14 ± 0.01	0.23 ± 0.01
ADNI [18]	Ours	0.72 ± 0.01	0.66 ± 0.02	0.09 ± 0.01	0.20 ± 0.02
PULSE whole-brain T1w	AE-UNet [27]	0.34 ± 0.02	0.28 ± 0.02	0.40 ± 0.01	0.51 ± 0.03
PULSE whole-brain T1w	DDPM [16]	0.38 ± 0.02	0.32 ± 0.02	0.35 ± 0.01	0.48 ± 0.02
PULSE whole-brain T1w	ANDi [12]	0.42 ± 0.03	0.36 ± 0.02	0.31 ± 0.02	0.44 ± 0.02
PULSE whole-brain T1w	Ours	0.53 ± 0.02	0.47 ± 0.02	0.22 ± 0.01	0.34 ± 0.01
PULSE brainstem T1w	AE-UNet [27]	0.35 ± 0.01	0.30 ± 0.02	0.39 ± 0.02	0.50 ± 0.01
PULSE brainstem T1w	DDPM [16]	0.40 ± 0.02	0.35 ± 0.02	0.34 ± 0.01	0.45 ± 0.02
PULSE brainstem T1w	ANDi [12]	0.50 ± 0.02	0.44 ± 0.02	0.27 ± 0.02	0.40 ± 0.02
PULSE brainstem T1w	Ours	0.62 ± 0.02	0.57 ± 0.02	0.18 ± 0.01	0.30 ± 0.01
PULSE brainstem T1w+FA	AE-UNet [27]	0.37 ± 0.02	0.32 ± 0.02	0.38 ± 0.01	0.49 ± 0.01
PULSE brainstem T1w+FA	DDPM [16]	0.41 ± 0.01	0.36 ± 0.02	0.33 ± 0.02	0.44 ± 0.01
PULSE brainstem T1w+FA	ANDi [12]	0.51 ± 0.01	0.46 ± 0.02	0.26 ± 0.02	0.38 ± 0.01
PULSE brainstem T1w+FA	Ours	0.65 ± 0.01	0.60 ± 0.01	0.16 ± 0.01	0.29 ± 0.01

6.4 Ablation and reliability analysis

Table 5 shows that final behavior arises from residual diversity, uncertainty-aware scoring, and healthy-reference calibration rather than simple thresholding. Residual-only projection yields BraTS/MSLesSeg Max-IoU of 0.34/0.24 and small-lesion Recall@5% of 0.43; equal-weight multi-view fusion improves these to 0.39/0.28 and 0.50, but raises healthy/stress FPRs to 0.318/0.392, amplifying non-pathological deviations as well as lesion evidence.

Healthy-reference fusion with LCB scoring achieves the strongest Max-IoU of 0.48/0.37, retains small-lesion Recall@5% at 0.62, and reduces healthy FPR to 0.074. The final FPR-controlled variant slightly lowers Max-IoU to 0.46/0.35, retains recall at 0.62, and reduces healthy/stress FPRs to 0.047/0.086, reflecting healthy-reference false-positive control rather than maximum overlap alone.

This ablation uses visible lesion masks and controlled healthy perturbations to quantify component effects. For mask-free ADNI and PULSE, interpretation instead uses ROI concentration, reliable ROI concentration, FP@Non-ROI, and unrelated ROI control, supporting S_{final} as calibrated, hypothesis-generating evidence rather than supervised segmentation or voxel-wise ALS pathology.

Table 5: Ablation of reliability components. Variants are evaluated by lesion localization, small-lesion recall, and healthy/stress FPR across residual-to-calibrated evidence stages; the calibrated operating point uses $\alpha = 0.05$.

Variant	Reliability / selection rule	BraTS Max-IoU $\uparrow$	MSLesSeg Max-IoU $\uparrow$	Small Recall@5% $\uparrow$	Healthy FPR $\downarrow$	Stress FPR $\downarrow$
Residual-only projection	R_{px}	0.34 ± 0.02	0.24 ± 0.02	0.43 ± 0.02	0.286 ± 0.021	0.361 ± 0.028
Equal-weight multi-view fusion	$\frac{1}{ \mathcal{M} \mathcal{S} } \sum_m R_m^{(s)}$	0.39 ± 0.01	0.28 ± 0.02	0.50 ± 0.02	0.318 ± 0.024	0.392 ± 0.030
Healthy-reference fusion + LCB	$S_{\text{LCB}}(v) = [\hat{R}(v) - \lambda_U U(v)]_+$	0.48 ± 0.01	0.37 ± 0.01	0.62 ± 0.01	0.074 ± 0.008	0.122 ± 0.011
FPR-controlled LCB selective	$S_{\text{LCB}} > q_{0.95}, U \leq \tau_U$	0.46 ± 0.01	0.35 ± 0.01	0.62 ± 0.01	0.047 ± 0.012	0.086 ± 0.019

7 Discussion and Conclusion

We presented *Reliable Normal Projection with Healthy-Reference Calibration*, a reliability-oriented framework for anomaly evidence localization in annotation-scarce brain MRI. Motivated by ALS brainstem analysis, where bulbar and respiratory dysfunction lack established voxel-wise MRI ground truth, we formulate the task as calibrated normal-projection evidence localization rather than supervised disease segmentation. Patient-to-healthy residuals should not be interpreted as anomaly evidence by default, since they may reflect disease-related change, anatomical variability, acquisition differences, preprocessing mismatch, low signal-to-noise ratio, partial-volume effects, or projection instability. Our framework addresses this ambiguity using multi-view residual cues, projection-disagreement uncertainty, and FPR calibration on held-out healthy reference scans.

The evaluation combines regimes with different reference information. Visible lesion datasets allow mask-based localization assessment, whereas ADNI and PULSE require ROI-centric interpretation because voxel-wise disease masks are unavailable. In PULSE, selected evidence should be read as hypothesis-generating deviation evidence within clinically motivated brainstem regions, not as voxel-wise ALS pathology or clinical validation. Brainstem-focused inference, fractional anisotropy when available, and unrelated ROI controls improve anatomical specificity and suggest that evidence concentration is not explained solely by brainstem-focused cropping. Nevertheless, ROI concentration measures anatomical plausibility and off-target suppression, not proof of disease specificity or clinical utility.

Several limitations remain. First, the PULSE evaluation is limited by the absence of voxel-wise disease ground truth and clinical-association analysis. Although ALSFRS-R, SNIP, and NIV status are available, relating calibrated evidence to respiratory or bulbar impairment, longitudinal progression, survival-related outcomes, and neurophysiological markers requires future validation. Second, healthy-reference calibration uses OpenBHB rather than dataset-matched healthy controls. Although OpenBHB covers a broad age range, it is skewed toward younger adults and does not fully represent late-life anatomical variability; thus, healthy-test false-positive rates reflect control on held-out OpenBHB scans, not guaranteed rates under every downstream age, scanner, site, or acquisition distribution. Third, leave-one-site-out calibration and large-scale multi-center transfer analysis remain future work, together with age-matched false-positive rates, dataset-matched calibration cohorts, age-aware normal projection, and larger older-control reference cohorts.

Clinically, calibrated evidence should be validated against longitudinal ALS progression, respiratory and bulbar measures, and independent neurophysiological markers before being considered as a biomarker. Overall, healthy-reference calibrated normal projection reframes anomaly detection as selective evidence calibration under annotation scarcity, providing a reliability-oriented protocol for normal-projection medical imaging backbones when dense labels are unavailable but false-positive control, calibrated evidence, and robustness remain essential.

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Compliance with Ethical Standards

OpenBHB, BraTS, and MSLesSeg are publicly available research datasets, while ADNI data are available to approved researchers under the ADNI Data Use Agreement. PULSE is a prospective observational French multicenter study performed in line with the Declaration of Helsinki and sponsored by the University Hospital of Lille (ID RCB 2013-A00969-36, ClinicalTrials.gov NCT02360891), with funding from ARSLA.

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